AI-Enabled Cultural Tourism in Sichuan: The Bashu Cultural Narrative Chain and Intelligent Resource Matching

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Abstract. Sichuan's cultural tourism sector faces overlapping challenges, including peak-season visitor surges, mountainous transportation bottlenecks, and governance constraints in ecologically sensitive areas. Under these conditions, conventional checklist-style scenic-spot recommendations often fail to translate into executable itineraries. This paper proposes the "Bashu Cultural Narrative Chain" as an experiential backbone. Using cultural symbols as thematic threads, it connects interpretation, experience, service, and communication across the pre-arrival, on-site, and post-departure phases, while incorporating carrying capacity, time windows, and risk as system inputs. Building on this structure, the paper advances an intelligent matching approach oriented toward destination operations: structuring people, places, events, objects, arts, and flavors through knowledge graphs; balancing experience enhancement and governance constraints via multi-objective optimization; improving narration and multilingual service efficiency through AIGC; and embedding trust mechanisms—including fact verification, copyright compliance, and AI-generated content labeling—into the content generation and service workflow. Through a comparative analysis of three scenario types—hard-constraint management at Jiuzhaigou, content-driven urban night tours in Chengdu, and weekend micro-trips in the Chengdu–Chongqing metropolitan area—the paper demonstrates that intelligent matching must adjust the prioritization of content and constraints in response to context in order to be practically viable.

Keywords: artificial intelligence; smart tourism; Sichuan cultural tourism; knowledge graph; AIGC

1. Introduction: From Resource Abundance to Experience Accessibility

Sichuan's cultural tourism resources are abundant, yet what visitors perceive is a continuous journey: motivation formation, information-based decision-making, transportation arrival, in-trip experience, and post-trip sharing and revisitation. Any imbalance along this chain can turn high-quality resources into unsatisfactory experiences.

In practice, three sets of tensions are particularly salient. First, hotspot congestion versus underutilization of lesser-known destinations: visitor flows are captured by a small number of 'viral' attractions, driving up pressure on public services. Second, fragmented narratives versus incoherent experiences: cultural symbols are scattered across scenic areas, performances, dining, and creative retail, lacking a unifying thread. Third, peak-season carrying-capacity pressure versus off-season supply shortages: during peak periods, advance reservation and entry caps must be imposed, while off-seasons lack stable products and communication momentum [1]. Reservation-based entry caps at hotspots such as Jiuzhaigou remind us that once tickets and entry time windows become hard boundaries, recommendations that ignore constraints can only remain 'beautiful on paper' [2].

It is worth emphasizing that visitors' understanding of accessibility extends well beyond physical arrival. It also encompasses information accessibility, service accessibility, and experience accessibility: When is the best time to depart? Is advance reservation required? What alternatives exist on a rainy day? Will the itinerary be too strenuous for elderly family members or young children? These questions may seem trivial, yet they directly determine satisfaction—and they are precisely the aspects most easily overlooked by traditional recommendation systems.

1.1 Research Questions and Contributions

This paper focuses on an operationally actionable question: In a provincial-level destination like Sichuan—characterized by high resource density, strong scenario volatility, and hard governance constraints—how can AI be advanced from content generation and point-of-interest recommendation to itinerary orchestration and operational scheduling, in a way that enhances experience without breaching governance and compliance boundaries?

Addressing this question, the paper's main contributions are: (1) proposing the experiential structure of the "Bashu Cultural Narrative Chain," grounding cultural expression from copywriting into executable itineraries; (2) providing an intelligent matching pathway based on knowledge graphs, explicit constraints, and explainable outputs, upgrading the unit of recommendation from individual points to narrative chains;

and (3) through comparative analysis of three representative scenarios, articulating priority-adjustment principles for different contexts, preventing the system from falling into the trap of generic recommendations.

1.2 Research Methods and Data Sources

This paper takes typical cultural tourism destinations in Sichuan Province as its research setting and selects three cases—one each representing an urban scenario, a natural scenic-area scenario, and a cultural-heritage scenario. Research materials are primarily drawn from three types of public information: (i) provincial and municipal policies, plans, and related textual materials; (ii) scenic-area and transportation operational information (including opening hours, reservation rules, capacity-control announcements, shuttle schedules, and risk notices); and (iii) platform-disclosed information (ticketing and reservation entry points, event releases, and visitor reviews), with a focus on the most recent three years.

Methodologically, the study uses thematic analysis of policy texts to extract narrative units and resource elements for the "Bashu Cultural Narrative Chain"; combines operational information to derive computable constraint fields; and, on this basis, constructs a rule-based matching framework oriented to different visitor segments. Finally, the paper outputs illustrative results for three cases and presents comparative observations across dimensions such as accessibility, reservation conflicts, congestion risk, and proxy indicators of satisfaction. Rules and factual descriptions referenced in this paper primarily draw on publicly available policy documents, scenic-area announcements, and local standards, with reference indices provided at the end [3][4].

2. Conceptual Framework: The Bashu Cultural Narrative Chain

This paper proposes the "Bashu Cultural Narrative Chain," emphasizing that it is not a copywriting technique but a mode of experience organization. The narrative chain comprises five sequential links: (1) cultural symbols—what to convey; (2) scenario nodes—where to experience it; (3) service processes—how visitors are served; (4) communication and conversion—how experiences are documented and shared; and (5) industrial synergy—how peripheral and long-tail consumption is stimulated.

The value of the chain lies in its capacity to accommodate constraints. Sichuan’s mountainous road networks, weather volatility, ecological protection requirements, and holiday peak visitor flows dictate that itineraries must be self-consistent in both time and space. Only when the unit of recommendation is upgraded from a list of attractions to a chain-based itinerary can the system simultaneously handle cultural coherence, physical pacing, transportation time costs, carrying capacity, and risk alerts.

To make the chain more intuitive: for visitors, it resembles a well-chaptered travel book. Even within Chengdu alone, one could take urban lifestyle as the main thread—stringing together lanes, teahouses, and night tours—or take the arc from ancient Shu museums to contemporary design as the main thread—connecting exhibitions, creative products, and hands-on craft experiences. Different main threads lead to different node selections, pacing arrangements, and service requirements, and this is precisely the explanatory power the narrative chain must provide.

3. Technical Pathway: From "Recommend" to "Implement"

3.1 Data and Standards: Getting the Objects Right First

Cultural tourism data is multimodal and inconsistently scoped. If opening hours, ticketing rules, transportation accessibility, and intensity levels are not unified, the system cannot generate executable itineraries. At minimum, the data foundation should achieve: unified calibration; usable interfaces (integrating ticketing, transportation, meteorology, early warning systems, etc.); and continuous updates (temporary controls and event schedules can be written back). Standardization and platformization are practical levers: standards decompose smart tourism into verifiable deliverables, while platforms turn capability reuse into sustained supply [5].

From an engineering implementation perspective, it is advisable to proceed with a minimum viable approach: first select one city or one main thread (e.g., urban night tours, a museum cluster, or county-level micro-trips), and complete basic fields—opening hours, ticketing rules, access modes, walkability, bikeability, elderly-friendly facilities, restrooms, and supply points—before gradually layering in dynamic information such as visitor flow, queuing, weather, and road conditions. The benefit of this approach is that the system is oriented toward executability from the outset, rather than building a trove of content displays and retrofitting operational data later.

3.2 Knowledge Graphs: Bringing Cultural Semantics into the Algorithm

Relying solely on popularity and similarity ranking makes it difficult to explain why items are arranged in a particular order, and it tends to funnel visitor flows toward a few viral hotspots. Knowledge graphs can organize "people, places, events, objects, arts, and flavors" into a relational network, enabling recommendations that consider not only preferences but also narrative coherence within the same chapter [6].

In implementation, a combination of tagging systems, narrative-consistency constraints, and spatiotemporal and carrying-capacity constraints can be adopted: unifying theme, season, and age-appropriateness calibrations; explicitly incorporating ticketing time windows, transportation time costs, queuing thresholds, altitude acclimatization, and risk levels; and outputting explainable rationales along with alternative options [7].

To make interpretability truly comprehensible to users, the output format should match usage contexts. Rather than abstract tags, it is preferable to present itinerary cards—for example: why items are arranged this way, estimated duration for each segment, where key constraints originate, and which nodes are substitutable. When constraint changes (sold out, road closure, rainfall) trigger rerouting, the system should explain the reason for the adjustment and the narrative thread that has been preserved, preventing users from perceiving the change as arbitrary.

3.3 AIGC: An Efficiency Tool, but with Trust Front-Loaded

AIGC can rapidly generate layered interpretive content, multilingual guided tours, and personalized explanations. However, cultural tourism content involves historical facts, folklore, and copyrighted materials, and errors or infringements can be amplified quickly [8][9].

Therefore, it is recommended that AIGC be primarily used for communication conversion and service support, with three safeguards established: factual verification, copyright compliance, and AI-generated content labeling with risk interception. For contested local narratives, it is better to present sources and differing accounts than to deliver a single definitive conclusion.

3.4 Operational Closed Loop: Connecting Recommendations to Scheduling

From a systems engineering perspective, recommendation is merely the entry point; what truly determines experience quality is whether the operational chain can deliver: Is reservation smooth? Is shuttle service honored? Is queuing predictable? Is rerouting explained? Is complaint resolution closed-loop?

It is recommended to treat prediction, recommendation, reservation, on-site scheduling, and feedback review as the minimum closed-loop unit. The prediction side uses holidays, weather, and historical visitor flow as priors; the recommendation side outputs executable itinerary cards; the reservation side integrates with ticketing and event schedules; the on-site scheduling side provides alternative plans based on real-time visitor flow, closures, and emergencies; and the feedback side writes evaluations, complaints, and fulfillment data back into rules and models. Only when the closed loop is operational can AI move beyond merely ‘making content prettier.’

4. Representative Scenarios: Three Comparative Cases and Strategy Differentiation

For intelligent matching, different scenarios pose different primary questions: some prioritize constraints, some prioritize content, and some prioritize short-frequency trips

and activation of underutilized resources. Below are three representative Sichuan contexts.

To clarify how constraint handling and recommendation strategies differ across scenarios, the paper first summarizes commonly used hard-constraint fields in intelligent matching, together with their public sources and roles, as shown in Table 4-1.

Table 4-1. Hard-Constraint Fields and Data Sources

Field Category	Example Fields	Public Data Sources	Role in Matching
Time window	Opening hours / sessions / reservation time slots	Scenic-area announcements; reservation systems	Filter infeasible combinations
Capacity	Daily maximum carrying capacity / time-sliced capacity caps	Scenic-area announcements	Control congestion risk
Transportation	Shuttle schedules / travel time	Public transit; traffic authorities; map platforms	Compute accessibility and travel cost
Risk	Weather alerts / closure notices	Meteorological agencies; emergency management; scenic-area announcements	Safety constraints and alternatives
Rules	Advance-booking days / real-name requirements	Reservation-platform descriptions	Penalize conflicts and ensure feasibility

4.1 Jiuzhaigou and Other Hotspot Scenic Areas: Constraints First

The core challenge of hotspot scenic areas lies in boundary conditions: ticket sell-outs, entry time windows, shuttle capacity, boardwalk capacity, and ecological red lines collectively determine whether a trip can be completed. The carrying-capacity limits and reservation mechanisms at scenic areas such as Jiuzhaigou exemplify this constraint logic [2].

A three-step approach is therefore recommended: first filter (ticketing, road conditions, weather, safety, etc.), then orchestrate (narrative consistency and physical pacing), then schedule (provide alternative routes or alternative experiences in response to real-time visitor flow and risk changes, with clear explanations for adjustments).

4.2 Chengdu Night Tour (Jinjiang Corridor): Content First

Chengdu’s night-tour carrying capacity is relatively stable, and operations can achieve soft scheduling through performance frequency, activity time slots, and district distribution. The challenge lies more in content and pacing.

Taking the Jinjiang night tour as an example, visitor decision-making is more lightweight and socially oriented, with greater sensitivity to atmosphere, stopping points, and photo spots. The system should design the night tour as a walkable chapter: using short stories to connect waterfronts, bridges, docks, lanes, and dining-performance venues, with rest and pause points designed as rhythm markers. Simultaneously, visitor-flow distribution should inform crowd dispersal to avoid localized congestion.

AIGC is better suited here for interpretation and service support: writing the same route in different versions—such as family trips, friend outings, and first-time visits—and clearly explaining why specific stops merit extended engagement.

4.3 Weekend Small-Town Micro-Trips in the Chengdu–Chongqing Metropolitan Area: Short-Frequency Travel and Activation of Underutilized Resources

Weekend micro-trips in the Chengdu–Chongqing metropolitan area are characterized as "short, frequent, and light": using high-speed rail or intercity transit as the backbone, centered on a half-day to full-day lightweight tour of a small town. The pain point is not a lack of resources, but that resources are scattered and lack low-cost entry points. Old streets, riverside walkways, local museums, and intangible cultural heritage workshops often have no unified narrative, and visitors find it difficult to judge how to use limited time optimally.

This type of scenario is better suited for micro-itinerary orchestration. First, establish the itinerary skeleton around the return-trip boundary (treat the last return train as a hard constraint and work backward to determine usable time slots). Second, organize small-town resources into narratable chapters and embed underutilized resources into the main thread (e.g., using a structure of "local crafts, one street, one meal, one stretch of riverbank" to turn intangible heritage experiences or small exhibitions into memory anchors). Third, design the first hour after arrival well (first meal, first walking segment, first photo spot) to reduce decision costs.

A more actionable output template might look like this: upon arrival, arrange a short-distance, low-intensity opening segment (old street, riverside walkway), placing lunch and rest stops in the middle of the main thread. In the afternoon, arrange one reservable small experience (handicraft, guided tour, short performance), and reserve the last hour before return for free browsing and souvenir shopping. This template may seem plain, but

it places time boundaries, physical boundaries, and consumption nodes into a single itinerary card—making it far more acceptable to weekend visitors.

On the operations side, reservable small experiences can be bundled with arrival time slots to form light products usable upon arrival, with coupon packages, combo tickets, or roaming passes reducing trial-and-error costs. For the system, this represents the integration of recommendation and productization: recommendations must be able to land on "I just arrived at the station—where next, how to get there, and do I need a reservation?"

4.4 Scenario Comparison Summary: How Priorities Shift

Across all three scenarios, both narrative consistency and executability must be embedded in the same output. Differences lie in the dominant role of constraint intensity versus content supply.

Hotspot scenic areas prioritize constraints: the system must first ensure entry feasibility, trip completion, and explainability. Urban night tours prioritize content: the system must articulate pacing and memory anchors clearly, using soft scheduling for crowd dispersal. Metropolitan-area micro-trips prioritize time boundaries: the system must bind transit timetables, lightweight products, and activation of underutilized resources. This comparison reminds us not to conceive of the recommendation system as a single model, but rather as a multi-strategy ensemble that invokes different strategy templates under different scenarios.

5 Governance and Evaluation: Trustworthy, Controllable, and Assessable

When a recommendation system begins to influence visitor-flow distribution, transportation connectivity, emergency alerts, and complaint handling, it increasingly functions as part of the governance apparatus rather than merely a marketing tool. Baseline capabilities must include: data minimization and personal-information protection, content labeling, risk alerts, closed-loop complaint resolution, and traceable change records.

In governance practice, it is recommended that public-facing alerts be tiered: essential safety information (road closures, extreme weather, capacity sell-outs) should be placed at the top, with experience suggestions and optional alternatives following. At the same time, strategy replay capabilities should be retained for the management side to facilitate post-hoc review. Especially in cross-municipality coordination scenarios, responsibilities for issuing risk alerts, explaining rerouting rationales, and handling complaints should be clarified before the system goes live.

Evaluation can be organized into three indicator groups—experience, operations, and industry. The experience dimension examines satisfaction, NPS (Net Promoter Score), dwell time, revisitation rate, and complaint rate. The operations dimension examines congestion duration, queuing time, reservation fulfillment rate, and emergency response. The industry dimension examines average transaction value, conversion rate, off-season occupancy rate, and participation in creative and intangible heritage offerings. Methodologically, a baseline–threshold–A/B testing approach combined with event-based review can be employed, incorporating holiday and extreme weather scenarios into normalized iteration cycles.

6 Research Boundaries and Future Work

This paper provides a framework and pathway oriented toward practical implementation, but has not yet completed rigorous empirical validation. Future work needs to further address: institutional and technical coordination for cross-stakeholder data sharing; quantification and interpretability of cultural authenticity and appropriateness; and the cross-departmental collaboration costs and responsibility-chain design entailed by advancing from recommendation to scheduling.

From an implementation roadmap perspective, a phased approach of "single-point first, then cross-regional coordination" can be adopted: first establish a closed loop at a single scenic area or single city (recommendation → reservation → feedback), then expand to cross-scenic-area flow diversion and cross-municipality coordination; first achieve static executability (correct timing and rules), then achieve dynamic schedulability (responsiveness to visitor flow and risk changes). This pacing better aligns with the absorptive capacity of real-world governance and operations.

6.1 Policy and Practice Implications

From Sichuan’s three representative scenarios, the key contradiction in smart cultural tourism is not insufficient individual technologies, but how to build an executable, governable, and explainable system under complex constraints. Drawing on the framework and scenario comparisons presented above, this paper offers the following policy and practice implications.

(1) Advance destination-level systematic supply using the narrative chain as the organizing unit. Currently, many destinations still rely on fragmented management of individual scenic spots, projects, and events, leading to breakdowns in cultural expression and experience structure. Organizing resources into narrative chains helps unify cultural interpretation, scenario nodes, and service processes along a single itinerary thread, and provides stable structural units for intelligent matching. It is recommended that narrative-chain construction be front-loaded—before brand, content, and service design—in

destination product-system development, forming a continuously iterable chapter-based product library.

(2) Shift from application stacking to data and governance infrastructure [10]. Whether smart cultural tourism can operate sustainably often depends on whether calibration is unified, interfaces are usable, and update mechanisms are robust. Policy should direct more resources toward foundational field standards, cross-departmental data-use mechanisms, and the interoperability of public service systems for ticketing, transportation, meteorology, and early warning. Through platformization, common capabilities such as visitor-flow monitoring, queuing prediction, and risk alerting can be accumulated as a reusable foundation rather than one-off projects.

(3) Integrate smart cultural tourism into the public governance chain, rather than treating it solely as a marketing tool. When recommendations begin to influence visitor-flow distribution, ecological carrying capacity, and emergency response, they simultaneously bear governance consequences. In practice, hard-constraint information such as safety alerts, capacity sell-outs, and road closures should be placed at the highest priority level within recommendation strategies, and closed-loop mechanisms for rerouting explanations, complaint handling, and event review should be established. Institutionally, responsibility boundaries and publication authorities should be clarified, forming traceable strategy change records.

(4) Promote micro-itinerary productization in the metropolitan area to activate underutilized resources. Short-frequency travel in the Chengdu–Chongqing metropolitan area depends more heavily on light products usable upon arrival, and visitors are highly sensitive to time boundaries and decision costs. It is recommended that transportation hubs serve as entry points, organizing scattered resources—small-town old streets, riverside walkways, local museums, and intangible cultural heritage workshops—into half-day or full-day micro-itinerary templates, supplemented by reservable small experiences, coupon packages, or combo tickets to reduce trial-and-error costs. Through the narrative chain, underutilized resources can be embedded into the main thread, forming replicable weekend micro-trip products.

(5) Strengthen assessability in pilots to form a replicable pathway model. It is recommended that pilots be conducted at the level of a single scenic area, single urban district, or single route, with baseline indicators and thresholds clearly defined. Methods such as A/B testing and holiday-period reviews should be used to identify strategy effectiveness. On the basis of evaluation, strategies can be codified into templates and gradually expanded to cross-scenic-area flow diversion and cross-municipality coordination. This approach reduces systemic risk and better matches the realistic pace of cultural tourism governance.

7 Conclusion

The key to Sichuan's smart cultural tourism lies not in building yet another application, but in using the narrative chain to weave dispersed resources into journey structures that are comprehensible, executable, consumable, and governable—and then connecting recommendation, generation, scheduling, and feedback into a closed loop. The three-scenario comparison demonstrates that objectives may vary by context, but the baseline should remain consistent: executable, explainable, and governable. For academic research, the most valuable next step is to conduct data-driven closed-loop experiments and comparative evaluations on specific municipalities or routes, advancing the framework-level argument into testable empirical results.

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